

Capstone Project: Loan Default Prediction

Executive Summary

The models explored for predicting loan default applicants are Decision Tree, hyperparameter tuned Decision Tree, Random Forest, and hyperparameter tuned Random Forest. The performance metrics used for model evaluation are accuracy, precision, recall, f1 score. This evaluation method is also called confusion matrix and is used for performance of classification models.

The outcome of confusion matrix performance for Decision Tree model with default parameter is overfit model. The model performance is not good on test dataset with loan default (class 1) precision 59% and recall 59%. The precision metrics 59% implies that of all identified loan defaults, only 59% are correct (True positive/ (True positive + False positive)). The recall 59% implies that of all the loan defaults only 59% are accurately predicted (True Positive/ True positive + False negative).

Other models are applied to load default training and test data with incremental improvement in performance metrics. The Random Forest training dataset has the most improved results with loan default precision and recall. The Random Forest test data set loan default precision is 79% and recall is 64%. The improvement is not enough for models to be applied in banking and financial systems. The suggestion is to explore additional data analysis and data preprocessing before building models. For example, perform data preprocessing like converting numerical variable to categorical variable and binning categorical variable. For example: DEBTINC ratio could be converted to categorical variable with binning. The DEBTINC ratio can be binned as follows 1) Below 36%: Considered a good DTI ratio, indicating manageable debt relative to income, 2) 36% - 43%: Many lenders consider this acceptable, though closer scrutiny may be required, and 3) Above 43%: Often viewed as a high risk, making it harder to qualify for loans, particularly mortgages. Explore other methods to impute missing value such as KNN. Consider unsupervised learning models for data analysis and treating outliers.

The goal of developing a machine learning model for loan default with accuracy and efficiency using machine learning involves addressing data imbalance, conducting model evaluation. We have proposed a solution that has improved incrementally; however the model performance is not good and requires additional methodologies to process dataset.

Problem Summary

Identifying the default risk is important because borrower will not make timely interest payments and/or repay the principal. Defaulting on loan carries significant consequences, including legal action, credit score damage, and potential asset seizure. A consequence of default is the degradation of a country or entity's credit rating, making future borrowing more difficult and expensive. Additionally, it can potentially trigger a financial crisis, affect economic growth. The risky lending algorithms caused the 2008 mortgage crisis. Holistic analysis of borrower validity helps avoid mishaps like global recession caused by subprime mortgages defaulting. Today, the loan prediction algorithms are based on limited amount of data, like income, credit

score, recent job history, and heuristics. Identifying valid borrowers requires analyzing multiple aspects of borrower such as income, job stability, transaction history, and now looking at data from social media. The data provided for loan default prediction needs to be treated for example the existing data is numerical. This data can be converted to categorical variable and binning can be done to simplify the dataset to build machine learning model.

Overview of steps

Following steps were performed in this analysis.

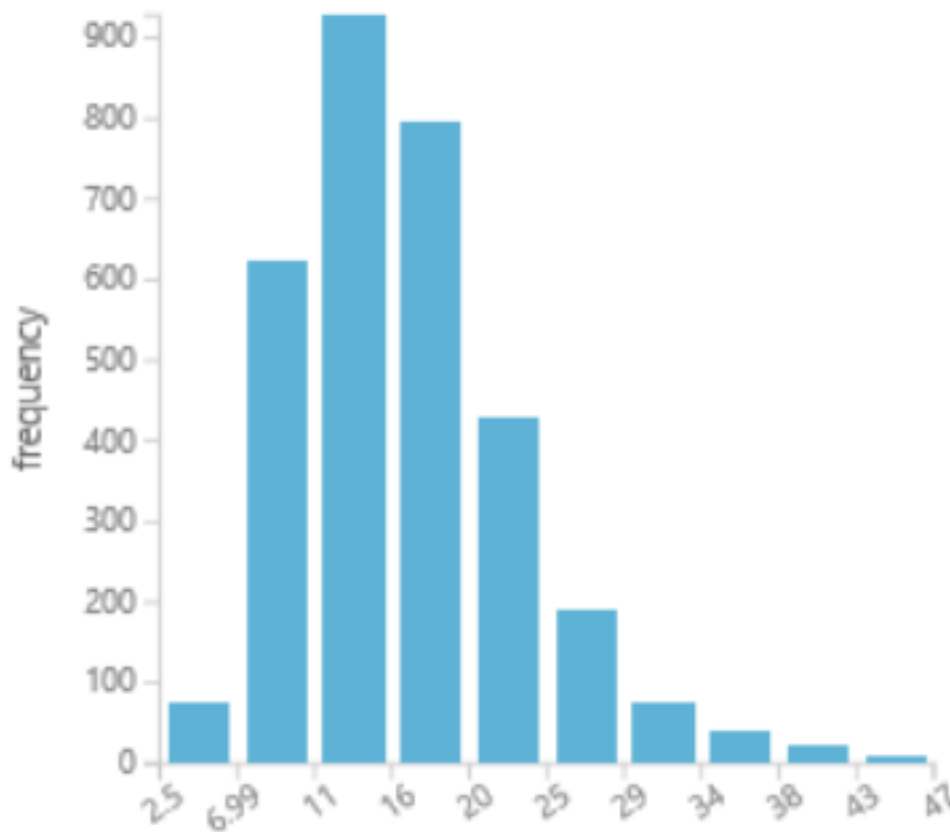
1. Data exploration
2. Data preparation and handling missing data
3. Feature selection, scaling, and editing Meta data
4. Regression modeling
5. Permutation Feature importance and feature engineering
6. Model evaluation and updates
7. Tuning model hyper parameters
8. Predicting loan default rate for private test data
9. Proposing further Analysis

Initial Data Exploration

Initial data exploration began with summary and descriptive statistics. Summary statistics for Loan Default rate is as follows.

Mean	16.8171
Median	15.8
Min	2.5
Max	47.4
Standard Deviation	6.698
Unique Values	341
Missing Values	0
Feature Type	Numeric Feature

Following histogram of poverty rate indicates that it has a uniform right tailed distribution.



Solution Design

For loan default prediction we explored classification machine learning models such as Decision Tree, hyperparameter tuned Decision Tree, Random Forest and hyperparameter tuned Random Forest machine learning model. The best model based on performance metrics is Random Forest model with default features. The performance metrics for Random Forest includes precision, recall and f1 score. The random forest loan default (or class 1) precision is 0.78, i.e. all the people identified with loan default the actual people with default loan are 78%. The loan default recall is 0.64, i.e. all the people who defaulted the correct prediction is 64%. The aim is to improve the loan default recall to higher than 90%. Improvement in recall metrics will identify high risk loan defaulters accurately and protect the financial institution from losing revenue. The loan default recall of 64% will lead to bank losing revenue on bad loans. If this model is applied overall by the country it can lead to financial system failure further causing recession.

After data preprocessing the most important features identified by machine learning model are DEBTINC, DELINQ, CLAGE. I'd like to explore methods that will identify DEROG, NINQ, CLNO and YOJ also as important features. I used median to impute missing value. I'd like to explore other imputing methods after additional research on important feature to include

DEBTINC, DELINQ, DEROG, CLAGE, CLNO, NINQ, and YOJ as the most important features. I'd drop REASON_HomeImp, Loan, MORTDUE and VALUE from the existing dataset to decrease the processing time for the tree and random forest.

Once the random forest loan default recall metric result is closer to 85%, I'd like to try boosting models such as Adaboost, Gradient boost, XG boost and ensemble machine learning models to improve the performance of model. It is advised that initial model test result should be good if not ideal for further improvement using hyperparameter tuning and advanced machine learning models. Use of hyperparameter tuning and advanced algorithm like random forest or boosting will improve the model incrementally. If the baseline model is not providing good performance, we need to go back to preprocessing the data to prepare for building the machine learning model.

Figure 1 shows the best model, which yields the confusion matrix test data result for loan default (class 1), precision 78% and recall 64%. Figure 2 shows the second-best model, the hyperparameter fine-tuned random forest model with confusion matrix test data result for loan default (class 2), precision 71% and recall 65%.

Figure: Confusion Matrix performance metrics for “Loan default test data”: Random Forest

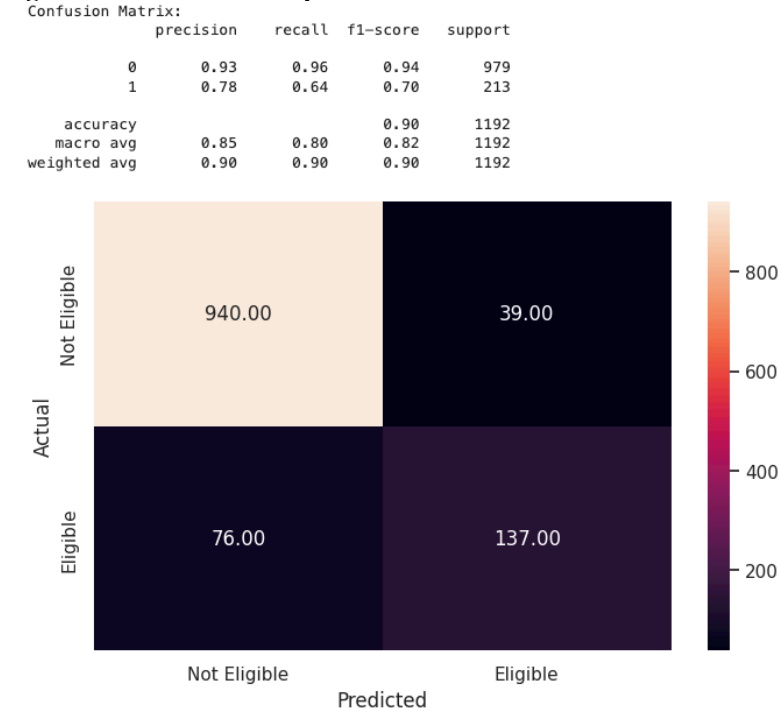
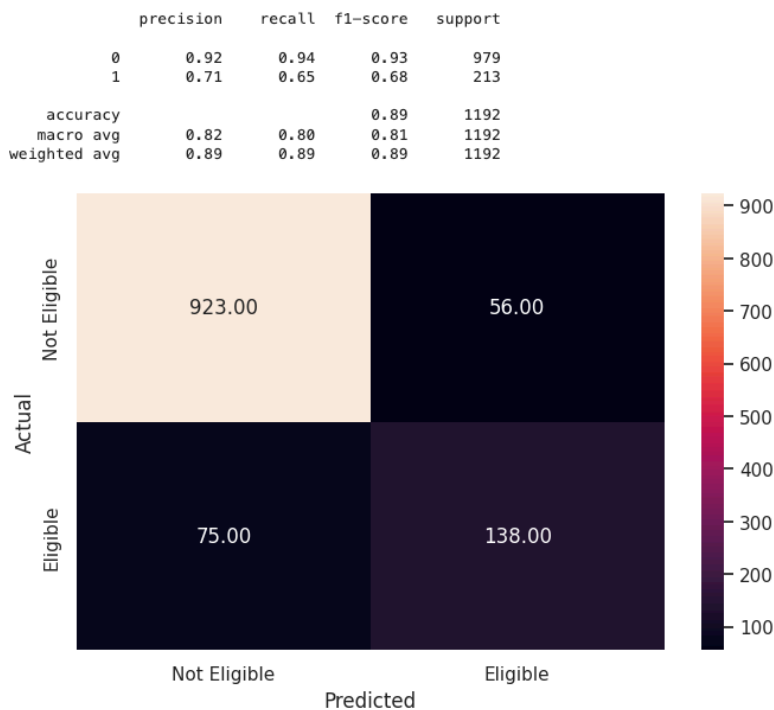


Figure: Confusion Matrix performance metrics for “Loan default test data”: Random Forest Tuned



Analysis and Key Insights:

The baseline machine learning model performance of loan default (class 1) recall values is not good. We performed hyperparameter tuning and applied tuned feature to Decision Tree for improved performance metrics. We used Random Forest and fine-tuned Random Forest for loan default prediction and evaluate the performance. The best performance metrics is for Random Forest with default features as shown in figure 2.

We analyze feature importance in Figure 4 for further insights. Based on feature importance analysis we must explore additional data processing before applying the dataset to machine learning models. We can explore KNN, and unsupervised learning to increase the importance of features like DEROG, YOJ, NINQ. In the existing dataset we can drop, VALUE, MORTDUE, and REASON from the dataset.

Figure: Confusion Matrix performance metrics for “Loan default test data”: Decision Tree

The loan default (class 1) precision is 59% and recall is 59%. This implies that data preprocessing needs more work

Confusion Matrix:

	precision	recall	f1-score	support
0	0.91	0.91	0.91	979
1	0.59	0.59	0.59	213
accuracy			0.85	1192
macro avg	0.75	0.75	0.75	1192
weighted avg	0.85	0.85	0.85	1192

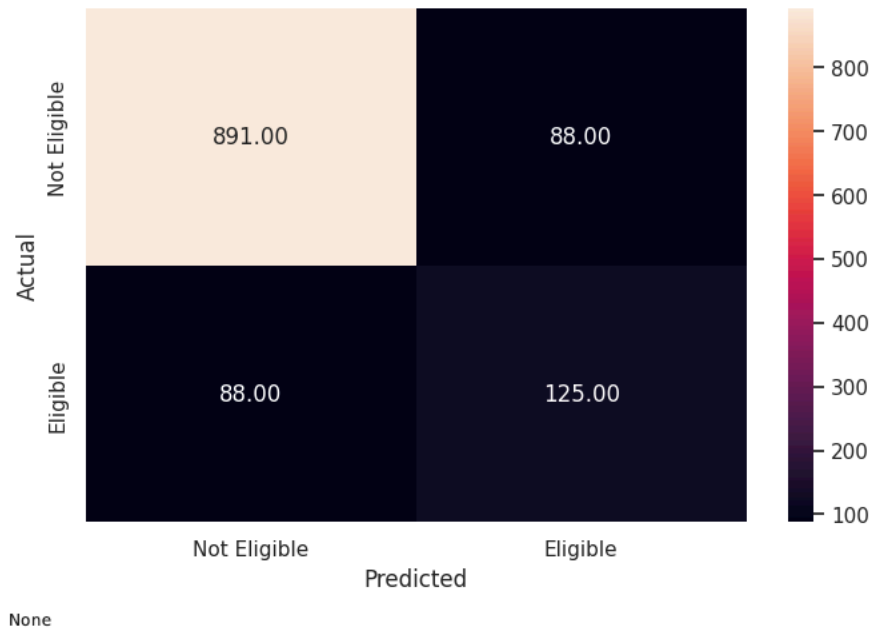
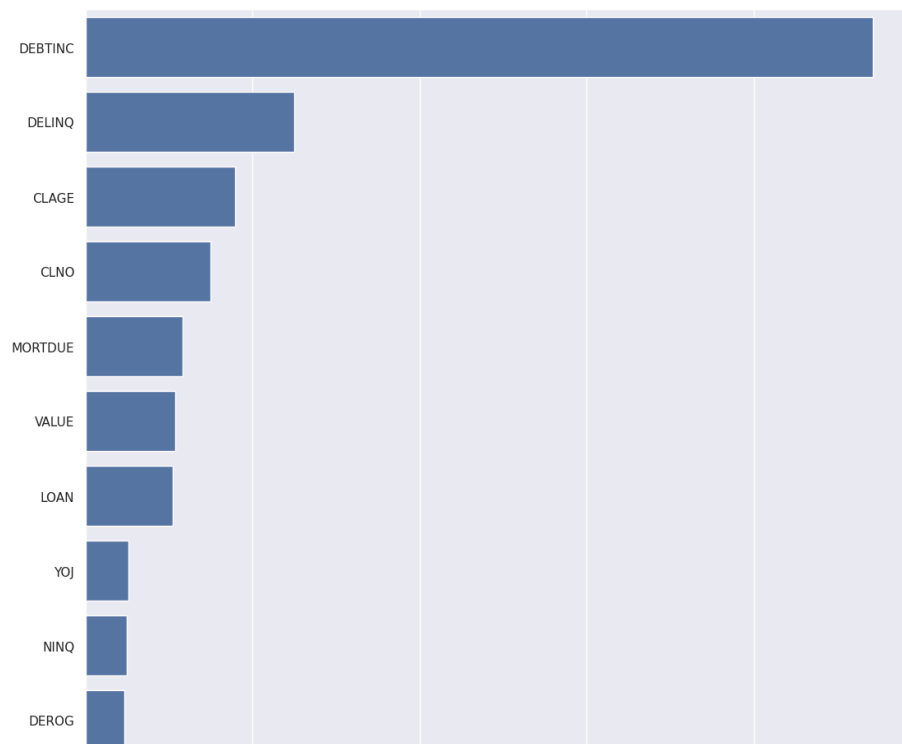


Figure: Feature importance used for building the tree

The features used below are DEBTINC, DELINQ and CLAGE as most important features. The features DEROG, NINQ, CLAGE, CLNO and YOJ should be of same importance as DELINQ and CLAGE. The following features should be dropped VALUE, REASON, MORTDUE to reduce the processing time for faster execution of fine-tuned random forest.



Limitation and Recommendation for Further Analysis

The proposed Decision Tree and Random Forest model performance metrics does not qualify for the models to be used by financial institutes and banks to identify loan defaults. The data preprocessing must include features like NINQ, CLNO, YOJ, DEROG to build the decision tree at depth 2,3 and 4. The existing model used median to impute the missing values. I'd like to research other methods to increase the weight of other features. I'd also like to experiment with separate set of features with people with less YOJ, higher NINQ and create a model with stringent requirement for loan approval. This leads to 2 models one with people with higher CLAGE, NINQ and less YOJ and other with people less CLAGE, NINQ and more YOJ.

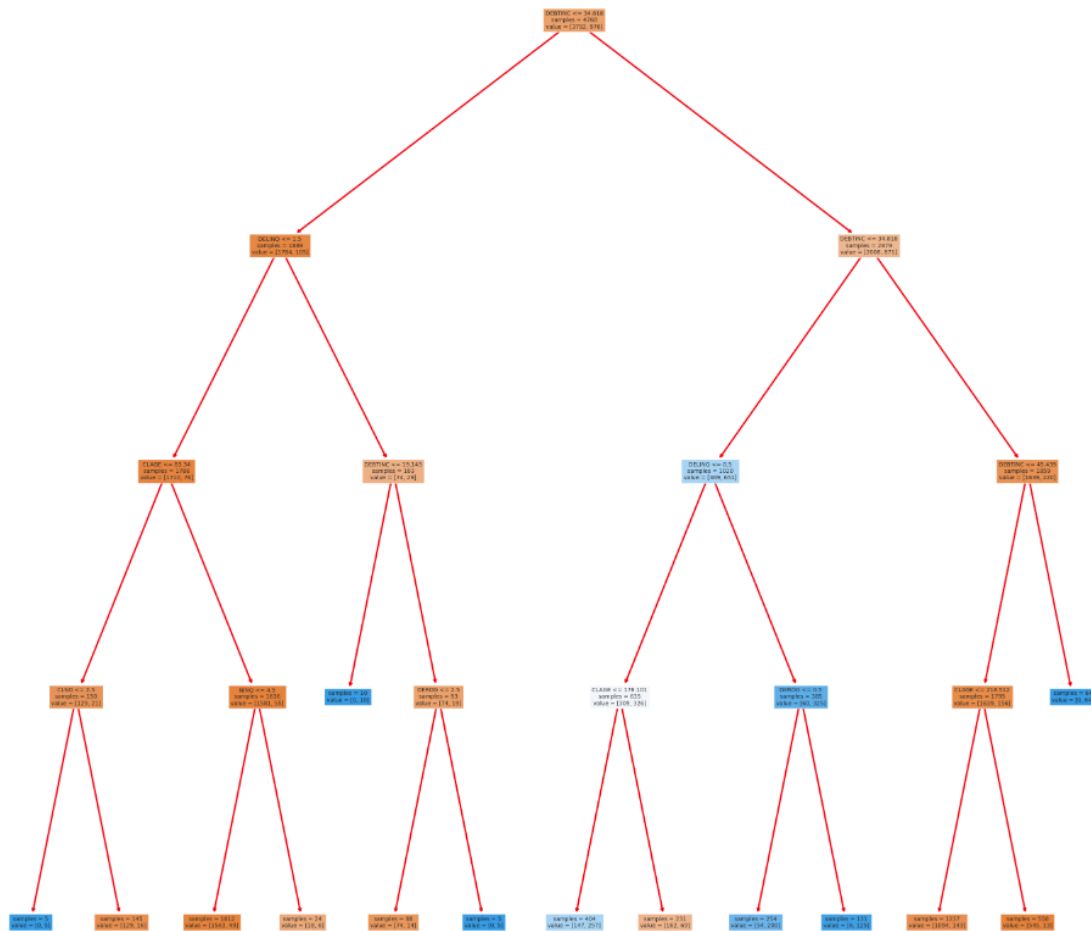
Figure 5: displays the dataset. The datatype of dataset is mostly numerical. I'd like to convert the dataset into categorical during data processing.

Figure: Loan default data set data type

```
df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5960 entries, 0 to 5959
Data columns (total 24 columns):
 #   Column              Non-Null Count  Dtype
---  -
0    BAD                  5960 non-null   category
1    LOAN                 5960 non-null   int64
2    MORTDUE              5960 non-null   float64
3    VALUE                5960 non-null   float64
4    REASON               5960 non-null   category
5    JOB                  5960 non-null   category
6    YOJ                  5960 non-null   float64
7    DEROG                5960 non-null   float64
8    DELINQ               5960 non-null   float64
9    CLAGE                5960 non-null   float64
10   NINQ                 5960 non-null   float64
11   CLNO                 5960 non-null   float64
12   DEBTINC              5960 non-null   float64
```

Figure below visualizes the decision tree. The root node is DEBTINC since DEBTINC is the most important feature. Following DEBTINC is DELINQ and CLAGE, CLNO. The depth of tree is 4.

Figure: Decision Tree with root node DEBTINC, tree depth is 5. Blue nodes indicate BAD LOAN.



Conclusion

Our process included univariant, bivariant data analysis, data pre-processing, data splitting into training and test data set. Subsequently 2 machine learning algorithms were tested DT and RF. The hyperparameter tuning was performed for both DT and RF and executed improved DT and RF. The models' performance is evaluated in terms accuracy, precision, recall and F1 score. Several features including DEBTINC, DELINQ, DEROG, CLAGE, CLNO, NINQ, VALUE, LOAN, REASON, MORTDUE, YOJ are considered for determining the BAD loan.

The proposed models did not show enhanced improvements. I'd explore more data balancing techniques such as KNN etc instead of median. Some features should be dropped or to enhance the applicability of the banl loan prediction model the furture should encompass the collection of more diverse and extensive real-life data, wider range of balancing approaches, exploration of unsupervised learning methods and utilization of ensemble techniques for building more reliable models for banking and financial institutions.

References

https://www.sciencedirect.com/science/article/pii/S2666307423000293?ssrnid=4376481&dgcid=SSRN_redirect_SD